Project Proposal

Project Title: PL Insights DWBI (DL3004 Final Project)
Domain: Sports Analytics & Business Intelligence
Repository: pitch side

Team Members:

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1. Executive Summary

PL Insights is an end-to-end Data Warehouse and Business Intelligence solution designed to analyze 16 Premier League seasons (2010/11 to 2025/26). The project ingests match telemetry, team performance statistics, referee assignments, and venue details to deliver analytical insights into competitive dynamics, home advantage, referee strictness, and tactical shot efficiency.

2. Business Justification & Goals

Modern sports enterprises require historical data consolidation to identify underlying performance patterns.

- **Goal 1:** Ingest 16 Premier League seasons (~6,080 matches / 12,160 fact rows).
- **Goal 2:** Build a star-schema Data Warehouse in PostgreSQL across 3 schemas (`project_staging`, `project_dw`, `project_reporting`).
- **Goal 3:** Perform advanced SQL OLAP analytics (ROLLUP, CUBE, GROUPING SETS).
- **Goal 4:** Deliver interactive executive dashboards in Power BI Desktop.